

# Le Quoc Viet

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## Professional Summary

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Electrical Engineering graduate with an Outstanding classification in Instrumentation and Automation from Óbuda University, Budapest, Hungary. Currently working as a Junior Embedded Software Developer at FPT Software while pursuing an M.Sc. in ACMI at Hanoi University of Science and Technology. Concurrently involved in research activities with the Military Technical Academy and the Signal, Information and Multimedia Content Processing Laboratory (SigM Lab) at HUST. Experienced in embedded and IoT system development across application software, middleware, low-level firmware, and hardware integration. Particularly interested in embedded systems for automotive, robotics, and autonomous aerial platforms, with a focus on real-time control, sensing, communication, and intelligent system integration.

## Professional Experiences

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**Laboratory Assistant at KVK Faculty of Óbuda University** 2024-2026

- Description:

Key Role: Documentation preparation and Teaching assistance in Theory and Laboratory Session for courses in Embedded Systems (AVR C and 32-bit systems).

Supporting Role: Development and presentation of a computer vision-based prototype for future Embedded Systems Laboratory Sessions, using Raspberry Pi 4B, Python, Picamera2, MediaPipe, and OpenCV for real-time hand-pose tracking and PySerial communication to control a 6-DOF Braccio robotic arm.

**Junior Embedded Developer at FPT Software** 2026-Current

- Description: Developed features and tests for MCAL layer (AUTOSAR) for MEM, FEE, MEMACC modules; Executing UT, CT, UM/IM Documentation and covering the coding standards in MISRA, CERT-C and HIS.

## Projects

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### Bare-Metal MCU and Device Driver Framework

- Designed a reusable bare-metal driver framework for the STM32L432KC using direct register-level programming and CMSIS definitions, providing hardware abstraction without relying on STM32 HAL peripheral APIs.
- Implemented configurable register-level drivers for communication peripherals (UART/USART/LPUART, I2C, SPI, CAN, SAI, USB), timing peripherals (TIM, RTC, LPTIM), system-control peripherals, ADC/DAC, and DMA, with interrupt-driven operation and low-power mode integration.
- Developed higher-level device drivers and event-driven state machines for IMU and ToF sensing, RFID and fingerprint-based authentication, and OLED/TFT display interfaces over designed peripherals layer.

### Raspberry Pi Edge Gateway and IoT Device Management Platform

- Architected a Python-based IoT edge platform using Raspberry Pi 4 as the local gateway and ESP8266-based remote sensor nodes, acquiring DHT11, IMU, and MQ-series sensor data through GPIO, I2C/SPI, and ADC interfaces and securely transmitting telemetry over Wi-Fi using MQTT over TLS. Implemented finite-state machines and an event-driven superloop for non-blocking scheduling, connection management, and sensor acquisition.
- Developed the Raspberry Pi gateway using a Mosquitto MQTT broker and Python services to process incoming telemetry, persist timestamped sensor data in SQLite, track device states, and perform anomaly detection. Exposed RESTful APIs through FastAPI for mobile access to device status, measurements, statistics, alerts, and remote commands.

### Pixhawk-Based UAV Platform

- Contributed to the physical system design and component selection of a four-rotor UAV, defining the component placement and wiring architecture, using a carbon-fiber quadcopter frame, four BDUAV 5010-360KV brushless motors, four Hobbywing XRotor Pro 50A ESCs, a 6S 22.2 V 9000 mAh LiPo battery, PM07 power-distribution board, Cube/Pixhawk flight controller, GPS, telemetry module, and FlySky radio-control system.
- Configured the flight-control system using Mission Planner, including IMU calibration, radio-controller setup,

GPS-assisted navigation, telemetry communication, and PID tuning for the quadcopter flight-control loops.

## Education

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**VietNam National University, Hanoi, Vietnam**, BSc in Data & Computer Science Sept 2021 – Sept 2022

- GPA: 3.22/4
- **Coursework:** Instructional & DSA in Java Programming, Shell Script in Ubuntu, Database (Microsoft SQL Server), MATLAB, Discrete Mathematics, Calculus 1,2 & 3, Linear Algebra.

**Óbuda University, Budapest, Hungary**, BSc in Electrical Engineering Sept 2022 – February 2026

- GPA: 4.06/5.0
- **Coursework:** Embedded C, C++ , Instructional Programming & Data Structures and Algorithms (C#, Python), STM32, Raspberry Pi 4, AVR C (ATmega64/128), Database Systems (MySQL), PCB Design, Verilog, SystemVerilog
- Graduated with Highest Honors: Outstanding (Final Degree Grade: 5/5).

**Hanoi University of Science and Technology , Hanoi, Vietnam**, MSc in Computer Science April 2026 – Current

## Awards

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**Stipendium Hungaricum Scholarship** 2022 - 2026

- Awarded the Hungarian Government's most prestigious higher education scholarship, with benefits often including tuition fees, accommodation, and medical insurance.

**TDK (Tudományos Diákkör) - Hungarian for Scientific Students' Association** 2024 - 2025

- Awarded 3rd Prize (2025)

### Multi-Technology Presence Management System

- Designed and implemented a distributed indoor/outdoor presence-detection system for BLE-emitting electronic cigarette devices, using BLE advertisement fingerprinting based on service/manufacturer data, RSSI characteristics, and temporal advertising patterns to classify target-device signatures beyond fixed MAC-address matching. Developed a multi-node RSSI fusion algorithm to estimate the most likely room/zone and assign a confidence score to each presence decision.
- Implemented the system on ESP32-S3-based Heltec LoRa32 v3 nodes with concurrent BLE scanning, local feature processing, and secure telemetry reporting over Wi-Fi/MQTT over TLS and LoRaWAN. Coordinated sensing and communication using FreeRTOS and low-power operation, with MATLAB-based analysis used to evaluate detection behavior, RSSI distributions, localization confidence, and communication performance.